Ades Mega Game - Application Summary

Hexello + Hexplode (internal mode id: hexpand)

Expanded rules, examples, and gameplay flow

What the app is

? A 3D hex-grid strategy game with Red vs Blue players.

? Each side can be Human or Computer, configured on the title screen.

? Two modes are available: Hexello (capture mode) and Hexplode (chain reaction mode).

? Board sizes: Hexello small/large = 4 or 6 per side, Hexplode small/large = 3 or 4 per side.

Core controls

? Tap a tile to play a move.

? One-finger drag rotates the board (yaw/pitch).

? Pinch zooms camera distance.

? Two-finger drag pans the board position.

Turn handling and input lock

The board accepts human taps only when intro animation is complete, the game is not over, no move animation is running, no AI move is queued, and the current player is not AI-controlled.

Why this matters

This prevents the human from taking extra turns while the AI is thinking or moving.

Human/AI turn starts ▶ **Move validates + applies** ▶ **Turn advances**

If next player is AI, minimax/heuristic selection runs, then AI move is applied.

Hexello Rules and Examples

Hexello is Reversi-style capture on a hexagonal board. You can only place on empty tiles that trap opponent pieces between the newly placed piece and one of your existing pieces along any of the six directions.

Rules

1. Board starts with a seeded center pattern: radius-1 ring alternates red/blue; center tile is randomized red or blue.
2. Current player is set to the opposite color of the randomized center tile.
3. A legal move must be on an empty tile and must produce at least one capture line.
4. When a move is played, all bracketed opponent tiles in each valid direction are flipped.
5. If a player has no legal move, turn passes with message "No Red Move Possible" or "No Blue Move Possible".
6. If both players have no legal moves, message "No Moves Possible" is shown and game ends by tile-count winner.

Example A (single line capture)

Suppose along one line you have R B B . . If Red places on the first empty dot that brackets the blue chain, the line becomes R R R R .

Diagram: Hexello flip line

Before: R B B . After red move: R R R R

Practical Hexello tips

? Corners are extremely strong and heavily favored in Hard AI evaluation.

? Avoid moves adjacent to empty corners unless you gain immediate tactical advantage.

? Mobility (number of legal moves) matters, not only immediate tile count.

Hexplode (Hexpand) Rules and Examples

Hexplode is a chain-reaction control game. You can play on empty cells or your own cells. Levels rise, then explode at level 6 (greater than 5), capturing and incrementing neighbors, which can trigger further explosions.

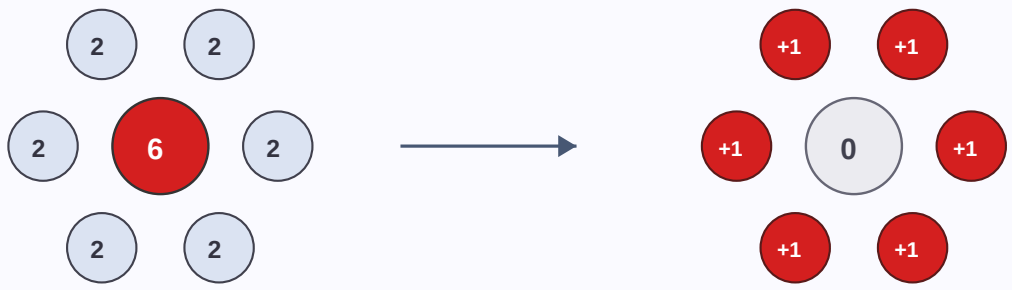
Rules

1. Legal move for player P: any cell where owner is empty or owner is P.
2. Applying move: owner = P, level += 1.
3. If level > 5, cell is queued for explosion.
4. Explosion processing: exploding cell resets to empty level 0; each neighbor becomes owned by exploding player and gets level +1.
5. Any neighbor that becomes level > 5 is queued, allowing chain reactions.
6. Game can end by elimination (after at least two total turns and both sides have appeared), and AI simulations also treat high score thresholds as terminal states.

Example B (chain reaction)

Your red level-5 cell is increased to level 6. It explodes, sending +1 to all neighbors. Two neighbors also cross to 6 and explode next, rapidly converting a region to red.

Diagram: Hexplode center blast and capture



Center resets to 0; each neighbor is captured/incremented; any that exceed 5 also explode.

AI behavior notes

? Easy mode uses simplified heuristics and intentional randomness/suboptimal play.

? Hard mode uses minimax with alpha-beta pruning and move ordering.

? Current hard depth for Hexplode is configured at 5 in code.

? Large boards have more legal moves and state branching, so full tree coverage is lower at same depth.